Biospectra : Vol. 10(2), Sept., 2015, pp 103-106
An International Biannual Refereed Journal of Life Sciences

Plant Science

ISSN : 0973-7057

Studies of xylem elements in *Euphorbia nivulia* Buch-Ham. in Maharashtra state of India

K. A. More*
Department of Botany, YashvantraoChavan Arts &Science Mahavidyalaya
Mangrulpirdist- washim (MH) India

Received : 11th July, 2015; Revised : 22nd August, 2015

Abstract : Xylem element studies in *Euphorbia nivulia* in Maharashtra State are described. The range of length and width of vessel elements was determined by the measurement of vessel elements and were classified as per the classification given by Radlford *et al.* (1974), which is reproduced here for further investigation. Various parameters of 30-50 xylem elements of *E. nivulia* surved show substantial variation in there morphological & numerical organisations. The details have been discussed in this paper.

Keywords : *Euphorbia nivulia*, xylem element, Percentage and range of length, Classification.

INTRODUCTION

Xylem is the tissue in vascular plants which conducts water (and substances dissolved in it) upwards in a plant. There are two kinds of cell which are involved in the actual transport: tracheids and vessel elements. Vessel elements are the building blocks of vessels, which constitute the major part of the water transporting system in those plants in which they occur. Vessels form an efficient system for transporting water (including necessary minerals) from the root to the leaves and other parts of the plant.

The presence of vessels in xylem has been considered to be one of the key innovations that led to the success of the flowering plants. It was once thought that vessel elements were an evolutionary innovation of flowering plants, but their absence from some basal *angiosperms* and their presence in some members of the *Gymnosperms* suggest that this hypothesis must be re-examined, vessel elements in some species of order *Gnetales* may not be homologous with those of angiosperms, or vessel elements that originated in a precursor to the angiosperms may have been subsequently lost in some basal lineages, So detail study of these xylem elements in Angiosperms required .

Now a day the xylem elements size, shape, orientation, structure of perforation plate and other characters are may be used to solve complicated issue related to taxonomic related problems of the plants.

*Corresponding author :
Phone : 09689081542
E-mail : kmore1914@gmail.com

MATERIALS AND METHODS

For study of vessels the preserved material were made into small pieces and boiled and cooled repeatedly until free from the air. A macerated fluid was prepared by taking aqueous chromic acid (as per Jeffrey's). The pieces of wood were kept in the fluid for 24 hours and after 24 hours the material was crushed with the help of glass rod and washed with distilled water to remove excess stain. The material was stained in 1 % saffranin for 6

hours and microscopic observations. The camera Lucida of the vessels were drawn by taking measurements the illustrations were drawn with India ink and microphotographs were taken wherever possible.

OBSERVATIONS

The various characters of vessel elements viz,.size wall thickening, shape, tail and characters of perforation plate like number, orientation and shape were observed. A survey of about 30-50 vessel elements of stem was carried out.

The range of length and width of vessel elements was determined by the measurement of 20-25 vessel elements and were classified as per the classification given by Radlford et al. (1974), which is reproduced here for persual.

A.	Extermely short	Less than 175 um
B.	Very short	175 to 250 um
C.	Moderately short	251 to 350 um
D.	Medium size	351 to 800 um
E.	Moderately Long	801 to 1100 um
F.	Very Long	over 1900 um

RESULTS AND DISCUSSION

Vessel element of *Euphorbia nivulia* Buch-Ham.

Vessel element of root (Table no-01)

Dimension :-

Extremely short (class A) very short (class B), moderately short (class C) and medium sized (class D) vessels were observed the frequency of very short (class A) was higher (39.22) and medium size (class D) shows lower frequency (10.78). The average diameter of vessels element is 19 mu.

Lateral wall thickening :-

simple pitted thickening were commonly observed, pits alternate.

Tail :-

Vessel with long pointed, long blunt, short pointed, short blunt are observed.

Perforation plate :-

In the vessel only simple perforation plate were present.

Orientation :-

The vessels with oval, oblique and transverse perforation plates were observed.

Shape of perforation plate :-

More commonly vessels have lenticular oval lin shape.

Root Fibers :-

The length of root fiber 422-617 mu.and average length of fibers is 576 mu. The width of fiber is between 9-18 mu. and average width is 9 mu. all the fibers are pointed at both the ends.

Tracheids :-

Average length of tracheid element is between 335-481 mu. and average length is 396 thewith of tracheids is between 9-18 mu. and average diameter is 108 mu all the tracheids are spindle shaped.

Vessel element of stem (Table no 02)

Euphorbia nivuliaBuch-Ham. **:-**

Very short (class B) moderately short (class C) medium sized (class D) vessels were commonly observed. The frequency of very short moderately short (class C) was higher (40.36) and frequency of very short vessel (29.54) shows lower frequency.

Shape :-

The shape of vessel element is cylindrical linear.

Lateral wall thickening :-

Simple pitted thickening were common pits alternate.

Tail :-

With long blunt, long pointed, short blunt were commonly observed.

Perforation Plate :-

In the vessel, only simple perforation plates were observed.

Orientation :-

The vessel with oblique and transverse perforation plate were observed.

Shape of perforation plate :-

More commonly vessels have perforation plate oval in shape.

Stem Fibers :-

The length of stem fiber is between 390-445 mu average length is 420 mu. and the width of fibre is in between 0.8-18 mu and average width is 0.9 mu all the fibers are pointed at both ends.

Tracheids :-

The length of tracheid element is in between 310-390 mu and average length 15-342 mu and the diameter of tracheid element is in between 16-21 mu. and average diameter is 18 mu. all the tracheids are spindle shaped.

Classification (After Radford etal) and relative frequency (%) of diferent classes of vessel element in the root and stem of *Euphorbia nivulia*Buch-Ham.

Table no 01 - Vessel elements of *Euphorbia nivulia* Buch-Ham. root shows percentage and range of length in each class

Class A		Class B		Class C		Class D	
Percentage %	Range of Length (mu)	Percentage %	Range of Length (mu)	Percentage %	Range of Length (mu)	Percentage %	Range of Length (mu)
39.22	72 to 144	30.00	180 to 234	20	252 to 260	10.78	410 to 450

Table no 02 - Vessel elements of *Euphorbia nivulia* Buch-Ham. stemshows percentage and range of length in each class

Class B		Class C		Class D	
Percentage %	Range of Length (mu)	Percentage %	Range of Length (mu)	Percentage %	Range of Length (mu)
29.54	180 to 216	40.36	262 to 324	30.10	360 to 522

CONCLUSION

Percentage and range of length (Dimention) is varies in root and stem of *Euphorbia nivulia* Buch-Ham.**But in*Euphorbia nivulia***Buch-HamShape, Lateral wall thickening ,Tail ,Perforation Plate ,Orientation ,Shape of perforation plate of vessels cant shows much more differentiation.

REFERANCES

1. **Anonymous(1968).** The preparation of wood for microscopicexamination. Forest Prod. Res. Lab. Leaflet No. 40.
2. **Bailey, I. W. (1953).** Evolutionofthetracheary tissue of land plants. Am.J. Bot.40:4—8.
3. **Bhandari M.M (1995).** Flora of the Indian deserts.
4. **Carlquist, S. 1984.** Vessel grouping in dicotyledon wood: significance and relationship to imperforatetracheary elements. Aliso 10: 505—525
5. **G. H. M. Lawrence (1968).** "Taxonomy of Vascular Plants", New York, the Macmillan Co., 1968, pp. 564-67
6. **IAWA Committee (1989).** IAWA list of microscopic features for hardwood identification. IAWA Bull. n.s.10: 219—332.
7. **Metcalfe, C.R. & L. Chalk**. (1983). Anatomy of the dicotyledons: wood structure and conclusion of the general introduction, 2. ed. Clarendon Press, Oxford. v. 2.
8. **Naik V. N. (1998)** Flora of Marathwada.
9. **W. W. Tupper, (1918).** Size variation in tracheary cells, I.A comparison between the secondary xylems of vascular Cryptogams, Gymnosperms and Angiosperms. Proc. Am. Acad. Arts Sci. 54:149—204
10. **Zimmermann, M.H. (1983).** Xylem structure and the ascent of sap.Springer – Verlag, Berlin.
11. **Radlford *et. al.*, (1978).** *Genera plantarum,* Oxford University Press, Londan.

* * *